STA2023 Exam 2 — Ch 4–6 — SIDE 1

■ BLUE = definition ■ **RED = formula / rule** ■ GREEN = example ■ BLACK = steps

SYMBOLS

Symbol	Means
P(E)	probability that event E happens (0 to 1)
N(E), N(S)	# of outcomes in E; # of outcomes in the whole sample space
E^c / "not E"	complement = everything that is NOT E
${}_nC_r$, ${}_nP_r$, n!	combinations (order doesn't matter), permutations (order matters), factorial
X / x	random variable / one specific value of it
P(x)	probability that X equals x
μ_X, σ_X	mean (expected value) and standard deviation of X
n, p, x	binomial: # of trials, P(success) on one trial, # of successes
z	z-score = how many SDs a value is from the mean
Σ	"add them all up"

Round probabilities to 4 decimals. Every answer: formula → plug in → arithmetic → boxed answer.

CH 4 — PROBABILITY

Basics

- **Probability** = long-run proportion of times an outcome happens. **0 ≤ P(E) ≤ 1; all outcomes add to 1.** **Unusual** = P < 0.05.
- **Sample space S** = all possible outcomes; **event** = some of them.
- **Classical** (equally likely): **P(E) = N(E) ÷ N(S).** **Empirical** (from data): **P(E) ≈ frequency ÷ total.** **Subjective** = educated guess.
- **Disjoint** = can't both happen (no overlap). **Independent** = one doesn't change the chance of the other.

NOT: $P(E^c) = 1 - P(E)$
OR: P(E or F) = P(E) + P(F) − P(E and F) (disjoint → P(E and F) = 0)
AND (independent): P(E and F) = P(E)·P(F)
AND (general): P(E and F) = P(E)·P(F | E)
GIVEN: $P(F \mid E) = \frac{P(E \text{ and } F)}{P(E)} = \frac{N(E \text{ and } F)}{N(E)}$
AT LEAST ONE: P(at least 1) = 1 − P(none)
Independent test: P(F | E) = P(F)? yes → independent

Contingency table (two-way table) ★Q1

1. Add row totals and column totals; grand total = N.
2. "is A" → (A's total) ÷ N.
3. "A **or** B" → P(A) + P(B) − P(both) (subtract the overlapping cell).
4. "A **and** B" → the one cell ÷ N.
5. "A **given** B" → cell ÷ (B's row or column total). Divide by the GIVEN group only.

Q: "Employment status of 78 residents: Full-time M 22 / F 18; Part-time M 6 / F 10; Unemployed M 3 / F 2; Not in labor force M 5 / F 12. Males = 36, Females = 42." Row totals: full-time 40, part-time 16, unemployed 5, not in labor force 17.
(a) P(male) = 36/78 = **0.4615**
(b) P(part-time) = 16/78 = **0.2051**
(c) P(male OR unemployed) = 36/78 + 5/78 − 3/78 = 38/78 = **0.4872**
(d) P(female AND not in labor force) = 12/78 = **0.1538**
(e) P(part-time | female) = 10/42 = **0.2381** (divide by females only)

Q: "P(speeder pulled over) = 0.8; P(ticket | pulled over) = 0.9. P(pulled over AND ticket)?" **A:** 0.8 × 0.9 = **0.72**
Q: "P(24-yr-old male survives the year) = 0.9986. P(at least one of 1000 dies)?" **A:** $1 - 0.9986^{1000} = 1 - 0.2464$ = **0.7536**
Q: "One card: P(king or diamond)?" **A:** 4/52 + 13/52 − 1/52 = 16/52 = **0.3077**

Q: "Coin flipped twice: sample space? P(two heads)? P(at least one tail)?" **A:** S = {HH, HT, TH, TT}; 1/4 = 0.25; 3/4 = 0.75.
Q: "Pick 2 of 4 friends (A, B, S, D): P(A and B)? P(B goes)?" **A:** S = {AB, AS, AD, BS, BD, SD} → 1/6; 3/6 = 0.5.
Q: "Is {A 0.40, B 0.20} a valid probability model?" **A:** No — sum 0.60 ≠ 1. "{A −0.10, …}" → No — negative.
Q: "200 people: 22 carpool, 5 walk. P(carpool)? Unusual to walk?" **A:** 22/200 = 0.11; walk 5/200 = 0.025 < 0.05 → **unusual**.
Q: "Die: P(6)? P(6 | even)?" **A:** 1/6; given even → {2,4,6} → 1/3 (changed → dependent).
Q: "Are 'widowed' and 'female' independent? P(widowed) = 15.0/271.7 = 0.0552, P(widowed | female) = 11.5/138.1 = 0.0833." **A:** 0.0833 ≠ 0.0552 → **dependent**.
Q: "Two 24-yr-old males both survive (0.9986 each)?" **A:** independent → 0.9986^2 = **0.9972**. Rule of thumb: sample < 5% of population → treat as independent.

Counting

Sequence of choices: multiply p·q·r…
n! = n(n−1)(n−2)⋯1, 0! = 1
Order matters: ${}_nP_r = \frac{n!}{(n-r)!}$ Order doesn't: ${}_nC_r = \frac{n!}{r!(n-r)!}$

3-letter codes: 26·26·26 = 17,576 · chair/vice/sec from 14: 14·13·12 = 2,184 · ${}_7P_5$ = 7!/2! = 2,520 · ${}_{10}P_3$ = 10·9·8 = 720 · ${}_4C_2$ = 24/(2·2) = 6 · ${}_{20}C_4$ = (20·19·18·17)/24 = 4,845
P(4 cards all match) = 13 / ${}_{52}C_4$ = 13/270,725 ≈ 0.00005

CH 5 — DISCRETE RANDOM VARIABLES

Probability distribution ★Q2, Q3

- **Discrete** = countable values (0, 1, 2 …); **continuous** = any value in an interval (measured).
- **Probability distribution** = every value of X with its P(x) (table, graph, or formula).

VALID if: (1) each 0 ≤ P(x) ≤ 1 and (2) Σ P(x) = 1
$\mu_X = \Sigma [x \cdot P(x)]$ $\sigma_X = \sqrt{\Sigma [(x-\mu)^2 \cdot P(x)]}$
shortcut: $\sigma_X = \sqrt{\Sigma [x^2 \cdot P(x)] - \mu^2}$

1. P(x) = frequency ÷ total (if given counts).
2. Mean: multiply each x by P(x), add.
3. SD: (x − μ)², times P(x), add, square root.
4. Graph: probability histogram — one bar centered on each x, height = P(x).

Q: "Is this a valid discrete probability distribution? x: 0,1,2,3,4 with P(x): 0.34, 0.21, 0.13, 0.04, 0.01."
A: each P(x) is between 0 and 1 ✓, but 0.34+0.21+0.13+0.04+0.01 = 0.73 ≠ 1 → **NOT valid** (probabilities must add to 1).

Q: "Games played in the Stanley Cup, X: x = 4 (freq 20), 5 (20), 6 (26), 7 (17). (a) Construct the probability distribution. (b) Graph it. (c) Compute and interpret the mean. (d) Compute the SD."
(a) total = 20+20+26+17 = 83 → P(4) = 20/83 = 0.2410, P(5) = 0.2410, P(6) = 26/83 = 0.3133, P(7) = 17/83 = 0.2048 (sum ≈ 1 ✓)
(b) bars at x = 4, 5, 6, 7 with heights 0.2410, 0.2410, 0.3133, 0.2048
(c) μ = 4(0.2410) + 5(0.2410) + 6(0.3133) + 7(0.2048) = 0.9640 + 1.2050 + 1.8798 + 1.4336 = **5.48 games** → over many series, the Cup lasts about 5.48 games on average
(d) (4−5.48)²(0.2410) = 2.1904(0.2410) = 0.5279
(5−5.48)²(0.2410) = 0.2304(0.2410) = 0.0555
(6−5.48)²(0.3133) = 0.2704(0.3133) = 0.0847
(7−5.48)²(0.2048) = 2.3104(0.2048) = 0.4732
σ² = 1.1413 → σ = √1.1413 ≈ **1.07 games**

Q: "Chuck-a-Luck profit: −$1 (0.5787), $1 (0.3472), $2 (0.0695), $3 (0.0046). Mean?" **A:** −0.5787 + 0.3472 + 0.1390 + 0.0138 = **−$0.08** → lose 8¢ per game on average. **Insurance:** $100(0.9986) + (−$9,900)(0.0014) = 99.86 − 13.86 = **$86** expected profit.

Binomial ★Q4

Binomial experiment = (1) fixed n trials, (2) independent, (3) two outcomes (success/failure), (4) same p every trial. X = # of successes.

$P(x) = {}_nC_x \cdot p^x \cdot (1-p)^{n-x}$ ${}_nC_x = \frac{n!}{x!(n-x)!}$
$\mu = n \cdot p$ $\sigma = \sqrt{n \cdot p \cdot (1-p)}$
Excel: =BINOM.DIST(x, n, p, FALSE) exactly x · TRUE = P(X ≤ x)

Words	Compute
exactly x	P(x)
fewer than k / less than k	P(0) + … + P(k − 1)
at most k / no more than k / ≤ k	P(0) + … + P(k)
more than k / > k	1 − [P(0) + … + P(k)]
at least k / ≥ k	1 − [P(0) + … + P(k − 1)]
between a and b (inclusive)	P(a) + … + P(b)

Q: "60% of U.S. women say the minimum driving age should be 18. Random sample of 10 women. (a) Exactly 3? (b) Fewer than 3? (c) Between 1 and 3 inclusive? (d) For a sample of 200, expected number and SD?"
n = 10, p = 0.60, 1 − p = 0.40
(a) $P(3) = {}_{10}C_3(0.6)^3(0.4)^7 = 120 \cdot 0.216 \cdot 0.0016384$ = **0.0425**
(b) P(X < 3) = P(0) + P(1) + P(2)
$P(0) = (0.4)^{10} = 0.0001$ · $P(1) = 10(0.6)(0.4)^9 = 0.0016$ · $P(2) = 45(0.36)(0.4)^8 = 0.0106$
= 0.0001 + 0.0016 + 0.0106 = **0.0123**
(c) P(1 ≤ X ≤ 3) = P(1) + P(2) + P(3) = 0.0016 + 0.0106 + 0.0425 = **0.0547**
(d) n = 200: μ = 200(0.60) = **120**; σ = √(200·0.60·0.40) = √48 = **6.93**

Q: "Binomial or not? (1) Roll a pair of dice 10 times, count 7s. (2) Select flights until 10 late ones are found. (3) Pick 4 of 30 students, count females."
A: (1) YES: n = 10, p = 6/36 = 1/6, x = 0…10. (2) NO: n not fixed. (3) NO: without replacement from 30, trials not independent (4/30 = 13% > 5%).

Q: "On-time rate 0.70, 4 flights. Build the distribution." **A:** $P(0) = (0.3)^4 = 0.0081$ · P(1) = 4(0.7)(0.3)³ = 0.0756 · P(2) = 6(0.49)(0.09) = 0.2646 · P(3) = 4(0.343)(0.3) = 0.4116 · $P(4) = (0.7)^4 = 0.2401$ (sum 1 ✓).
Q: "Free throws 95%, 3 shots: P(all 3)? P(less than 3)? mean & SD?" **A:** 0.95^3 = **0.8574**; 1 − 0.8574 = **0.1426**; μ = 3(0.95) = **2.85**; σ = √(3·0.95·0.05) = √0.1425 = **0.38**.
Q: "72% prefer chocolate, n = 10: exactly 8? fewer than 3? at least 3?" **A:** $45(0.72)^8(0.28)^2 = 0.2548$; P(0)+P(1)+P(2) = 0.0010; 1 − 0.0010 = 0.9990.
Q: "Chuck-a-Luck SD?" **A:** μ = −0.0787; Σ(x − μ)²P(x) = 0.4912 + 0.4040 + 0.3003 + 0.0436 = 1.2391 → σ = **$1.11**.

- **Discrete vs continuous:** number of pieces / colors / people → discrete; mass, diameter, time, temperature → continuous.
- **Expected value** E(X) = μ_X: "in the long run, average of X per trial."

STA2023 Exam 2 — Ch 4–6 — SIDE 2

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CH 6 — CONTINUOUS & NORMAL

Density curves & uniform ★Q5

- **Probability density function (pdf)** = a curve where the AREA under it over an interval = the probability X is in that interval; total area = 1; curve ≥ 0.
- **P(X = one exact value) = 0**, so P(a < X < b) = P(a ≤ X ≤ b).
- **Uniform on [a, b]** = flat rectangle: every interval of the same length is equally likely.

Uniform height = $\frac{1}{b-a}$ P(c ≤ X ≤ d) = $\frac{d-c}{b-a}$ = (length of piece) × height
"P% within the next t" → t = P × (b − a)

Q: "Daytime temperature X is uniformly distributed between 70°F and 100°F. (a) Sketch/describe the pdf. (b) P(75 ≤ X ≤ 85)? (c) P(X < 80)?"
(a) rectangle from 70 to 100, height 1/(100 − 70) = 1/30 (area = 30 × 1/30 = 1)
(b) (85 − 75)/30 = 10/30 = **0.3333**
(c) (80 − 70)/30 = 10/30 = **0.3333**

Q: "Friend is 0 to 30 min late, uniform. P(10 to 20 min late)? There is a 20% chance they arrive within the next ___ min." **A:** 10/30 = 0.3333; 0.20 × 30 = **6 min**. "Random number 0–10: P(exactly 4.35)?" **A:** 0. "P(between 2 and 5)?" **A:** 3/10 = 0.3.

Normal distribution

Normal = bell-shaped, symmetric, described by μ and σ. Properties: (1) symmetric about μ; (2) mean = median = mode = μ (highest point); (3) inflection points at μ ± σ; (4) total area = 1; (5) tails approach the axis but never touch; (6) Empirical Rule: 68% within 1σ, 95% within 2σ, 99.7% within 3σ; (7) area over an interval = proportion of the population in it = probability one random value is in it.

Standard normal Z = normal with μ = 0, σ = 1. Table V gives the area to the LEFT of z.

"Sketch the curve": draw a bell, write μ under the peak, tick marks at μ ± σ, μ ± 2σ, μ ± 3σ; mark the x value(s), shade the asked region, write the area in the shaded part.

$z = \frac{x-\mu}{\sigma}$ (round z to 2 decimals) x = μ + z·σ
LEFT of z: read table · RIGHT of z: 1 − (left) · BETWEEN: (left of z_2) − (left of z_1)
Negative z: area left of −z = 1 − (area left of +z)

1. Draw the curve, mark μ, shade the region asked.
2. Convert each x to z (2 decimals).
3. Look up the area to the LEFT of each z (table below).
4. Left → done. Right → 1 − area. Between → subtract the two lefts.

Q: "(a) Area to the left of z = −1.10. (b) Area between z = −0.60 and z = 1.45. (c) X normal with μ = 80, σ = 10; find P(70 < X < 95)."
(a) table: **0.1357**
(b) left of 1.45 = 0.9265; left of −0.60 = 0.2743 → 0.9265 − 0.2743 = **0.6522**
(c) z = (70 − 80)/10 = −1.00; z = (95 − 80)/10 = 1.50 → 0.9332 − 0.1587 = **0.7745**

Q: "Wechsler IQ: μ = 100, σ = 15. (a) P(score above 125)? (b) P(below 90)? (c) Proportion between 110 and 140? (d) What score is the 98th percentile?"
(a) z = (125 − 100)/15 = 1.67 → left = 0.9525 → right = 1 − 0.9525 = **0.0475**
(b) z = (90 − 100)/15 = −0.67 → **0.2514**
(c) z = (110−100)/15 = 0.67 → 0.7486; z = (140−100)/15 = 2.67 → 0.9962 → 0.9962 − 0.7486 = **0.2476**
(d) area to the left = 0.98 → closest table value 0.9798 → z = 2.05 → x = 100 + 2.05(15) = **130.75 ≈ 131**

Q: "SAT math: μ = 516, σ = 116. P(less than 500)? P(more than 600)? P(between 525 and 550)?"
z = −0.14 → **0.4443** · z = 0.72 → 1 − 0.7642 = **0.2358** · z = 0.08 and 0.29 → 0.6141 − 0.5319 = **0.0822**

Q: "Cholesterol μ = 180, σ = 36.2; P(above 200)?" **A:** z = (200 − 180)/36.2 = 0.55 → 1 − 0.7088 = **0.2912** (≈ 0.2903 with unrounded z).
Q: "Pregnancy μ = 266, σ = 16; P(259 days or less)?" **A:** z = (259 − 266)/16 = −0.44 → **0.3300**.
Q: "IQ μ = 100, σ = 15: P(outside 70 to 130)?" **A:** z = ±2.00 → 0.0228 + 0.0228 = **0.0455**. "P(exactly 100)?" → 0.

Find x from a percent / percentile ("backwards")

1. Draw & shade. Turn the question into an area to the LEFT (k-th percentile → 0.k; "top 10%" → left area 0.90; "middle 80%" → 0.10 and 0.90).
2. Find that area INSIDE the table (closest value); read z from the edges.
3. x = μ + z·σ.

Percentile	z	Percentile	z
5th	−1.645	95th	1.645
10th	−1.28	90th	1.28
20th	−0.84	80th	0.84
25th	−0.67	75th	0.67
2.5th	−1.96	97.5th	1.96
1st	−2.33	99th	2.33
2nd	−2.05	98th	2.05

Middle 80% ↔ z = ±1.28 · middle 90% ↔ ±1.645 · middle 95% ↔ ±1.96 · middle 99% ↔ ±2.575

Q: "Heights of 3-yr-old girls: μ = 38.72 in, σ = 3.17. (a) Height at the 20th percentile? (b) Heights that separate the middle 80%?"
(a) area 0.20 → z = −0.84 → x = 38.72 + (−0.84)(3.17) = **36.06 in**
(b) z = ±1.28 → 38.72 − 1.28(3.17) = **34.66** and 38.72 + 1.28(3.17) = **42.78 in**
Q: "P(shorter than 35 in)?" z = (35 − 38.72)/3.17 = −1.17 → **0.1210**

Q: "IQ μ = 100, σ = 15: score for the top 10%? bottom 5%? middle 90%?" **A:** top 10% → left area 0.90 → z = 1.28 → 100 + 1.28(15) = **119.2** · bottom 5% → z = −1.645 → **75.3** · middle 90% → z = ±1.645 → **75.3 to 124.7**.

Interpreting an area; assessing normality

- "Area to the right of 200 is 0.2903" means: (1) the **proportion** of the population above 200 is 0.2903, and (2) the **probability** a randomly selected person is above 200 is 0.2903.
- **Normal?** Histogram: bell-shaped, one peak, roughly symmetric, few outliers. **Normal probability plot:** points close to a straight line → approximately normal.
- **Q:** "Restaurant A wait times: tall on the left, long right tail. Restaurant B: symmetric hump. Which is normal?" **A:** B (bell-shaped); A is skewed right → not normal. "Greyhound times: plot points roughly on a line?" → approximately normal.

Standard normal table — area to the LEFT of z (entries are 0.xxxx; positive z; negative z → 1 − value)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	5000	5040	5080	5120	5160	5199	5239	5279	5319	5359
0.1	5398	5438	5478	5517	5557	5596	5636	5675	5714	5753
0.2	5793	5832	5871	5910	5948	5987	6026	6064	6103	6141
0.3	6179	6217	6255	6293	6331	6368	6406	6443	6480	6517
0.4	6554	6591	6628	6664	6700	6736	6772	6808	6844	6879
0.5	6915	6950	6985	7019	7054	7088	7123	7157	7190	7224
0.6	7257	7291	7324	7357	7389	7422	7454	7486	7517	7549
0.7	7580	7611	7642	7673	7704	7734	7764	7794	7823	7852
0.8	7881	7910	7939	7967	7995	8023	8051	8078	8106	8133
0.9	8159	8186	8212	8238	8264	8289	8315	8340	8365	8389
1.0	8413	8438	8461	8485	8508	8531	8554	8577	8599	8621
1.1	8643	8665	8686	8708	8729	8749	8770	8790	8810	8830
1.2	8849	8869	8888	8907	8925	8944	8962	8980	8997	9015
1.3	9032	9049	9066	9082	9099	9115	9131	9147	9162	9177
1.4	9192	9207	9222	9236	9251	9265	9279	9292	9306	9319
1.5	9332	9345	9357	9370	9382	9394	9406	9418	9429	9441
1.6	9452	9463	9474	9484	9495	9505	9515	9525	9535	9545
1.7	9554	9564	9573	9582	9591	9599	9608	9616	9625	9633
1.8	9641	9649	9656	9664	9671	9678	9686	9693	9699	9706
1.9	9713	9719	9726	9732	9738	9744	9750	9756	9761	9767
2.0	9772	9778	9783	9788	9793	9798	9803	9808	9812	9817
2.1	9821	9826	9830	9834	9838	9842	9846	9850	9854	9857
2.2	9861	9864	9868	9871	9875	9878	9881	9884	9887	9890
2.3	9893	9896	9898	9901	9904	9906	9909	9911	9913	9916
2.4	9918	9920	9922	9925	9927	9929	9931	9932	9934	9936
2.5	9938	9940	9941	9943	9945	9946	9948	9949	9951	9952
2.6	9953	9955	9956	9957	9959	9960	9961	9962	9963	9964
2.7	9965	9966	9967	9968	9969	9970	9971	9972	9973	9974
2.8	9974	9975	9976	9977	9977	9978	9979	9979	9980	9981
2.9	9981	9982	9982	9983	9984	9984	9985	9985	9986	9986
3.0	9987	9987	9987	9988	9988	9989	9989	9989	9990	9990
3.1	9990	9991	9991	9991	9992	9992	9992	9992	9993	9993
3.2	9993	9993	9994	9994	9994	9994	9994	9995	9995	9995
3.3	9995	9995	9995	9996	9996	9996	9996	9996	9996	9997
3.4	9997	9997	9997	9997	9997	9997	9997	9997	9997	9998

Reading the table: z = 1.67 → row **1.6**, column **.07** → 0.9525. Negative z: left of −1.67 = 1 − 0.9525 = 0.0475. **Backwards:** find the area inside the table (0.98 → 0.9798 is closest) → read z from the edges (row 2.0, column .05 → z = 2.05).

EXAM 1 SKILLS YOU STILL NEED

Mean: μ = Σx ÷ N (population) · $\bar{x}$ = Σx ÷ n (sample)
Median = middle of SORTED data (even n → average the two middles) · Mode = most frequent (none if no repeats)
Variance: σ^2 = Σ(x − μ)² ÷ N · s^2 = Σ(x − $\bar{x}$)² ÷ (n − 1) · SD = √variance
$z = \frac{x-\mu}{\sigma}$ (bigger z = relatively higher) · Relative frequency = count ÷ total (= empirical probability) · % = RF × 100
Quartiles: Q_2 = median; Q_1/Q_3 = medians of the halves (odd n → leave the median out) · IQR = Q_3 − Q_1 · fences Q_1 − 1.5·IQR, Q_3 + 1.5·IQR → outside = outlier
Empirical Rule (bell-shaped): 68% within 1σ, 95% within 2σ, 99.7% within 3σ; bands 34 | 13.5 | 2.35 | 0.15 each side (IQ 85–130 → 34+34+13.5 = 81.5%)

1. SD by hand: mean → each (x − mean) → square → add → ÷ N or ÷ (n − 1) → √.
2. Calculator 1-Var Stats: σx = population SD, Sx = sample SD; variance = (unrounded SD)².

Q: "Sample 18, 22, 25, 28, 30: mean and sample variance?" **A:** $\bar{x}$ = 123/5 = 24.6; squares 43.56 + 6.76 + 0.16 + 11.56 + 29.16 = 91.20 → s^2 = 91.20/4 = **22.8**, s = **4.77**.
Q: "ACT 27 (μ 18, σ 6) vs SAT 1200 (μ 1000, σ 200) — who did better?" **A:** z = 1.50 vs z = 1.00 → **ACT**.
Q: "Bell-shaped, μ = 29.8, σ = 6.4: 95% lie between?" **A:** 29.8 ± 2(6.4) → **17.0 to 42.6**; between μ+1σ and μ+2σ → 13.5%.

DON'T LOSE POINTS

- "Or" → subtract the overlap once. "And" → the cell. "Given" → divide by the given group, not the grand total.
- Check Σ P(x) = 1 before calling a distribution valid.
- Sample vs. whole: binomial μ = np, σ = √(np(1−p)) — don't forget the √.
- "Fewer than 3" stops at 2; "at least 3" = 1 − P(≤ 2); "at most" includes the number.
- Uniform: probability = (length ÷ total width); never use z here.
- Normal: ALWAYS convert to z (2 decimals) first; table = area to the LEFT.
- "Above / more than" → 1 − left. "Between" → big area minus small area.
- Percentile question → area → z (from inside the table) → x = μ + zσ.
- Round probabilities to 4 decimals; show the formula every time.

Which rule? (Ch 4)

- One event, equally likely → N(E)/N(S); from data → freq/total.
- "**OR**" → add, subtract overlap. "**AND**" → multiply (P(F | E) if dependent).
- "**GIVEN** / of those who…" → cell ÷ given group. "**AT LEAST ONE**" → 1 − P(none).
- Counting: sequence → multiply; order matters → ${}_nP_r$; doesn't → ${}_nC_r$.